



# Dr Ronan Dupont

## Curriculum Vitae

Nagoya, Japan

☎ (+33) 6 70 90 88 08

✉ [r-dupont@na.nuap.nagoya-u.ac.jp](mailto:r-dupont@na.nuap.nagoya-u.ac.jp)

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*Postdoctoral Researcher at Nagoya University (名古屋大学 博士研究員)*

## Education

- 2021–2024 **PhD, Applied Mathematics for Computational Physics**, *GeoSciences, IMAG, CNRS, Montpellier University*, Montpellier, France.  
Wave-morphodynamic coupling of the coastline by minimization principle. Under the supervision of Pr. Bijan Mohammadi and Pr. Frédéric Bouchette ([PDF](#)).
- 2018–2021 **Engineering degree in Computational Fluid Dynamics**, *SeaTech, Engineering School, Toulon University*, France.
- 2020–2021 **Master in Marine Sciences**, *University of Toulon*, France.  
Physics of the Environment for the Evaluation of Risks (PHYMER).
- 2016–2018 **Preparatory classes for French engineering schools**, *Caen, France*.

## Publications

### Journal Articles

- 2025 **Ronan Dupont**. An arbitrary-order virtual element method for the helmholtz equation applied to wave field calculation in port. *Results in Applied Mathematics*, volume 26, page 100598. Elsevier, 2025, ([PDF](#)).
- 2024 **Ronan Dupont**, Frédéric Bouchette, and Bijan Mohammadi. Beaches morphodynamic modeling based on hadamard sensitivity analysis. *Ocean Modelling*, volume 189, page 102370. Elsevier, 2024, ([PDF](#)).
- 2023 **Ronan Dupont**, Megan Cook, Frédéric Bouchette, Bijan Mohammadi, and Samuel Meul . Sandy beach dynamics by constrained wave energy minimization. *Ocean Modelling*, volume 183, page 102197. Elsevier, 2023, ([PDF](#)).

### In Conference Proceedings

- 2022 **Ronan Dupont**, Megan Cook, Frédéric Bouchette, Bijan Mohammadi, and Damien Sous. Optimorph: un mod le de morphodynamique du littoral par principe de minimisation. analyse de sensibilit  en 1d et application multi-1d. volume 17, pages 327–336. JNGCGC, 2022, ([PDF](#)).

## Research Experience

### Nagoya University 名古屋大学 (Japan 日本)

- 2026–Present **Postdoctoral Researcher in Numerical Linear Algebra**.  
Research on iterative methods for large sparse linear systems, including Krylov subspace methods. Development and analysis of numerical solvers for scientific computing applications.
- Supervisor: **Pr. Tomohiro SOGABE** (曾我部知広), expert in numerical linear algebra and Krylov subspace methods.

## Montpellier University (France)

### 2024 ***Solving the mild-slope equation using the Virtual Element Method (VEM).***

Virtual element method of order  $k$  with Robin's Boundary condition. Application to a concrete problem: the port of Cherbourg, ([PDF](#)).

Associate : **Dr. Mathias DAUPHIN**, *PhD on high-order numerical methods.*

## SeaTech, Engineering School, Toulon University (France)

### 2021 ***2D mesh of a sphere.***

Modeling the surface mesh (2D) of a sphere in Fortran 90 and display in Python. Create types and functions to number vertices and store their coordinates. Application to poisson problems.

Advisor : **Pr. Cédric GALUSINSKI**, *Professor, Department of Mathematics, IMATH, Toulon University* ([Personal Web-page](#)).

### 2021 ***Sudoku solving using a genetic algorithm method.***

Establishing the optimization method for a given problem.

Advisor : **Dr. Sylvain MAIRE**, *Associate Professor, Department of Mathematics, IMATH, Toulon University*

### 2021 ***Resolution of bi-fluid Euler equations, application with the BBAMR code.***

Resolution in Fortran 90. Solving by the Finite Volume method using different schemes (Godunov, HLLC, Lax) on Fortran.

Advisor : **Dr. Frederic GOLAY**, *Associate Professor, Department of Mathematics, IMATH, Toulon University* ([Personal Web-page](#)).

### 2020 ***Modeling the spread of epidemics in France using the SIR model.***

Modeling and solving systems of partial differential with diffusion in Python. Change from SIR to SZR model of zombie propagation.

Advisor : **Dr. Gloria FACCANONI**, *Associate Professor, Department of Mathematics, IMATH, Toulon University* ([Personal Web-page](#)).

## Segula Technologies, Trappes, France

### 2021 ***CFD optimization of the performance of windsurf sails intended for high speeds.***

6-month end-of-studies research internship.

Advisor : **M. Laurent LANQUETIN & Dr. Thibaut ALLEAU.**

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## Talks

### 2026 ***ICMASE 2026 - Nagoya (Japan).***

On September 14, 2026, I presented  $GPBiCG(m, \ell, k)$ : An Extension of  $GPBiCG(m, \ell)$  and Preliminary Adaptive Parameter Strategies and served as a **session chair** at this conference ([Talk page](#)), ([PDF](#)), ([Progressive PDF](#)). An associated Moonshot project poster is also available ([PDF](#)).

### 2024 ***PhD Defense - Montpellier (France).***

On September 30, 2024, I defended my thesis in Montpellier ([PDF](#)).

### 2024 ***NuMerics2024 : Numerical Methods for Problems in Fluid Dynamics - Naples (Italy).***

I was a **guest speaker** at this workshop and my work was entitled *Numerical solution of Mild-slope equation using Virtual Element Method* ([Conference website](#)), ([PDF](#)).

### 2023 ***AGU 2023 - San Francisco (United States).***

I presented a poster at this international conference showing *The generic version of the OptiMorph model* ([Conference website](#)), ([PDF](#)).

### 2023 ***Journées de Modélisation des Vagues à Phases Résolues - Île d'Aix (France).***

I took part in this workshop and presented *The OptiMorph model forced by a wave-to-wave resolution model (Shallow Water)* ([Conference website](#)), ([PDF](#)).

### 2022 ***Journée Nationales Génie Côtier - Génie Civil 2022 - Chatou (France).***

I took part in this conference and presented *OptiMorph: a coastal morphodynamics model based on the minimization principle. Sensitivity analysis in 1D and multi-1D application.* ([Conference website](#)), ([PDF](#)).

## Skills

Programming Python, Matlab, Fortran, C, C++,  $\LaTeX$ .  
Num Lin Alg Iterative methods, Krylov subspace methods, Preconditioning, Sparse solvers.  
CFD Software OpenFoam, Fluent, ADINA.  
Optimization Optimal transport, Genetic algorithms, Stochastic methods.  
HPC Bash, Slurm.  
Interests Numerical modelling, Fluid mechanics, Coastal physics, Finite Elements, Virtual Elements, Finite Volumes.

## Languages

French  Native language.  
English  Fluent - C1.  
Italian  Intermediate - B1.  
German  Pre-Intermediate - A2.  
Japanese  Pre-Intermediate - A2.  
Spanish  Basic - A1.

## Teaching Experience

2025 **Mathematics teacher at an international school**, *EFI French International School of Phnom Penh*, Phnom Penh, Cambodia.  
2025 **Private mathematics tutoring**, *Eurêka Tutoring Centre*, Phnom Penh, Cambodia.  
2025 **Secondary-school mathematics teacher**, *Ensemble scolaire Saint-Joseph de Tivoli*, Bordeaux, France.  
2023–2024 **Private tutoring: Algebra, Calculus, Cardinality, Geometry**, *First year of mathematics degree*, Montpellier University, France, ([Article mentioning my support for a top sportswoman](#)).  
2023 **Advanced coastal and port hydromorphodynamics**, *Master of Coastal Engineering*, Montpellier University, France, ([Course material](#)).  
2023 **Coastal and port hydromorphodynamic modeling tools: OptiMorph**, *Master of Coastal Engineering*, Montpellier University, France, ([Course material](#)).  
2023 **Python Courses**, *Master of Coastal Engineering*, Montpellier University, France, ([Course material](#)).

## Referees

### Pr. Tomohiro Sogabe

*Professor, Applied Physics*  
Graduate School of Engineering  
Nagoya University  
✉ sogabe@na.nuap.nagoya-u.ac.jp

### Pr. Bijan Mohammadi

*Professor, Department of Mathematics*  
Institut Montpelliérain Alexander Grothendieck  
✉ bijan.mohammadi@umontpellier.com

### Pr. Mehmet Ersoy

*Professor, Department of Mathematics*  
IMATH Toulon  
☎ +33 6 72 53 36 33  
✉ mehmet.ersoy@univ-tln.fr